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import cv2
import numpy as np
import matplotlib.pyplot as plt

# Create a test image
img = np.zeros((500, 700, 3), dtype=np.uint8)

# Draw shapes
cv2.rectangle(img, (80, 100), (250, 280), (255, 255, 255), 3)
cv2.circle(img, (380, 180), 80, (255, 255, 255), 3)

pts = np.array([[520, 300], [620, 100], [680, 300]], np.int32)
cv2.polylines(img, [pts], True, (255, 255, 255), 3)

cv2.line(img, (100, 400), (600, 400), (255, 255, 255), 3)
cv2.line(img, (250, 350), (500, 150), (255, 255, 255), 3)

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# ----- EDGE DETECTION -----
edges = cv2.Canny(gray, 50, 150)

# ----- HARRIS CORNER DETECTION -----
gray_float = np.float32(gray)

harris = cv2.cornerHarris(
    gray_float,
    blockSize=2,
    ksize=3,
    k=0.04
)

harris = cv2.dilate(harris, None)

harris_result = img.copy()
harris_result[harris > 0.01 * harris.max()] = [0, 0, 255]
```

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# ----- SHI-TOMASI CORNER DETECTION -----
shi_result = img.copy()

corners = cv2.goodFeaturesToTrack(
    gray,
    maxCorners=100,
    qualityLevel=0.01,
    minDistance=10
)

if corners is not None:
    corners = np.int32(corners)

    for corner in corners:
        x, y = corner.ravel()
        cv2.circle(
            shi_result,
            (x, y),
            5,
            (0, 255, 0),
            -1
        )

# ----- DISPLAY RESULTS -----
plt.figure(figsize=(15, 10))

plt.subplot(2, 3, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(edges, cmap="gray")
plt.title("Canny Edge Detection")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.imshow(cv2.cvtColor(harris_result, cv2.COLOR_BGR2RGB))
```

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plt.title("Harris Corner Detection")
plt.axis("off")

plt.subplot(2, 3, 4)
plt.imshow(cv2.cvtColor(shi_result, cv2.COLOR_BGR2RGB))
plt.title("Shi-Tomasi Corner Detection")
plt.axis("off")

# Combined result
combined = cv2.cvtColor(edges, cv2.COLOR_GRAY2BGR)

if corners is not None:
    for corner in corners:
        x, y = corner.ravel()
        cv2.circle(
            combined,
            (x, y),
            5,
            (0, 255, 0),
            -1
        )

plt.subplot(2, 3, 5)
plt.imshow(cv2.cvtColor(combined, cv2.COLOR_BGR2RGB))
plt.title("Edges + Corners")
plt.axis("off")

plt.tight_layout()
plt.show()

# ----- ANALYSIS -----
print("COMPARATIVE ANALYSIS")
print("=" * 50)

print("Edge Detection:")
print("- Detects object boundaries and outlines.")
print("- Useful for segmentation and shape detection.")

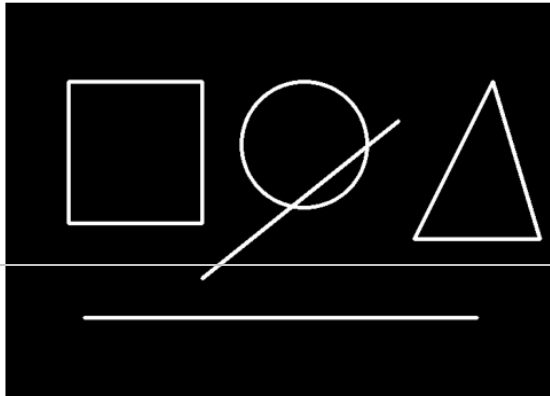
print("\nCorner Detection:")
```

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print("- Detects important points where edges meet.")
print("- Useful for feature matching and tracking.")

print("\nNumber of Shi-Tomasi Corners:",
      0 if corners is None else len(corners))

print("\nCONCLUSION:")
print("Edge detection captures boundaries, while corner detection")
print("captures distinctive feature points. Both are important")
print("for computer vision applications such as object recognition,")
print("tracking, image matching and image registration.")
```

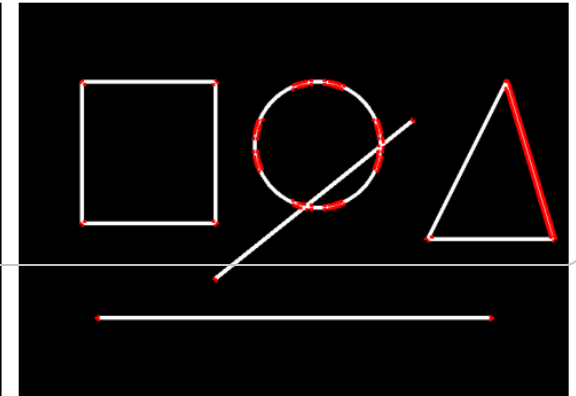
Original Image



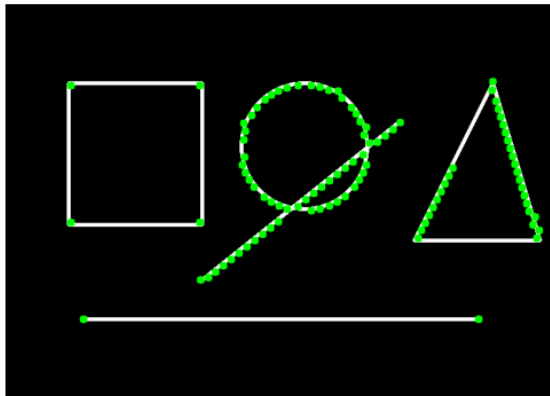
Canny Edge Detection



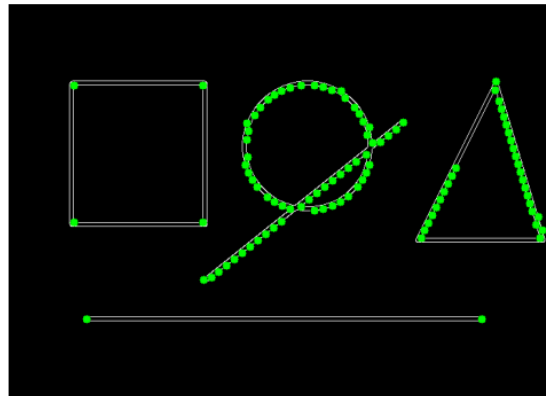
Harris Corner Detection



Shi-Tomasi Corner Detection



Edges + Corners



COMPARATIVE ANALYSIS

Edge Detection:

- Detects object boundaries and outlines.
- Useful for segmentation and shape detection.